

Nursery
INGREDIENT LAB ACTIVITY SYSTEM

IL-001 to IL-100

LOOK -> NAME -> MATCH -> SORT -> DRAW

Purpose	Build food and ingredient literacy by connecting plant structure, observation, measurement, post-harvest change, quality and product-development thinking.
Ingredient set	Hibiscus, ginger, turmeric, pineapple, citrus, mint, moringa, watermelon, tomato and maize.
Core rule	The Ingredient Lab is an educational investigation environment. It does not require tasting, medical claims or uncontrolled food processing.
Connections	Link each activity to GrowMeal vocabulary, journals, practical cards, Farm-to-Food activities, missions and assessment.

INGREDIENT LAB - HIBISCUS | PLANT PART: CALYX

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-001 - HIBISCUS - IDENTITY & PLANT PART

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-002 - HIBISCUS - EXTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-003 - HIBISCUS - INTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-004 - HIBISCUS - COLOUR & TEXTURE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-005 - HIBISCUS - MASS & SIZE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-006 - HIBISCUS - WATER/CHANGE OBSERVATION

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-007 - HIBISCUS - SORTING & QUALITY

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-008 - HIBISCUS - POST-HARVEST CHANGE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-009 - HIBISCUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-010 - HIBISCUS - INVESTIGATION CHALLENGE

Plant part	Calyx
Learning objective	Help children recognize hibiscus, identify its calyx as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the hibiscus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - GINGER | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-011 - GINGER - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-012 - GINGER - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-013 - GINGER - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-014 - GINGER - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-015 - GINGER - MASS & SIZE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-016 - GINGER - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-017 - GINGER - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-018 - GINGER - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-019 - GINGER - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-020 - GINGER - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Help children recognize ginger, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the ginger? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TURMERIC | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-021 - TURMERIC - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-022 - TURMERIC - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-023 - TURMERIC - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-024 - TURMERIC - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-025 - TURMERIC - MASS & SIZE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-026 - TURMERIC - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-027 - TURMERIC - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-028 - TURMERIC - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-029 - TURMERIC - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-030 - TURMERIC - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Help children recognize turmeric, identify its rhizome as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the turmeric? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - PINEAPPLE | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-031 - PINEAPPLE - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-032 - PINEAPPLE - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-033 - PINEAPPLE - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-034 - PINEAPPLE - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-035 - PINEAPPLE - MASS & SIZE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-036 - PINEAPPLE - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-037 - PINEAPPLE - SORTING & QUALITY

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-038 - PINEAPPLE - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-039 - PINEAPPLE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-040 - PINEAPPLE - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Help children recognize pineapple, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the pineapple? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - CITRUS | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-041 - CITRUS - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-042 - CITRUS - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-043 - CITRUS - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-044 - CITRUS - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-045 - CITRUS - MASS & SIZE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-046 - CITRUS - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-047 - CITRUS - SORTING & QUALITY

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-048 - CITRUS - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-049 - CITRUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-050 - CITRUS - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Help children recognize citrus, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the citrus? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MINT | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-051 - MINT - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-052 - MINT - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-053 - MINT - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-054 - MINT - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-055 - MINT - MASS & SIZE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-056 - MINT - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-057 - MINT - SORTING & QUALITY

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-058 - MINT - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-059 - MINT - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-060 - MINT - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Help children recognize mint, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the mint? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MORINGA | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-061 - MORINGA - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-062 - MORINGA - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-063 - MORINGA - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-064 - MORINGA - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-065 - MORINGA - MASS & SIZE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-066 - MORINGA - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-067 - MORINGA - SORTING & QUALITY

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-068 - MORINGA - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-069 - MORINGA - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-070 - MORINGA - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Help children recognize moringa, identify its leaf as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the moringa? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - WATERMELON | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-071 - WATERMELON - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-072 - WATERMELON - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-073 - WATERMELON - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-074 - WATERMELON - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-075 - WATERMELON - MASS & SIZE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-076 - WATERMELON - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-077 - WATERMELON - SORTING & QUALITY

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-078 - WATERMELON - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-079 - WATERMELON - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-080 - WATERMELON - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Help children recognize watermelon, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the watermelon? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TOMATO | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-081 - TOMATO - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-082 - TOMATO - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-083 - TOMATO - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-084 - TOMATO - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-085 - TOMATO - MASS & SIZE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-086 - TOMATO - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-087 - TOMATO - SORTING & QUALITY

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-088 - TOMATO - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-089 - TOMATO - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-090 - TOMATO - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Help children recognize tomato, identify its fruit as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the tomato? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MAIZE | PLANT PART: SEED/GRAIN

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-091 - MAIZE - IDENTITY & PLANT PART

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the identity & plant part task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-092 - MAIZE - EXTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the external structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-093 - MAIZE - INTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the internal structure task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-094 - MAIZE - COLOUR & TEXTURE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the colour & texture task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-095 - MAIZE - MASS & SIZE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the mass & size task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-096 - MAIZE - WATER/CHANGE OBSERVATION

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the water/change observation task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-097 - MAIZE - SORTING & QUALITY

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the sorting & quality task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-098 - MAIZE - POST-HARVEST CHANGE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the post-harvest change task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-099 - MAIZE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the ingredient-to-food pathway task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-100 - MAIZE - INVESTIGATION CHALLENGE

Plant part	Seed/Grain
Learning objective	Help children recognize maize, identify its seed/grain as the food-use plant part, and describe one visible feature.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Teacher shows a clean intact sample or picture. Children look, name, match, sort or draw according to the investigation challenge task. Use only teacher-prepared cut samples for visual observation.
Evidence to record	Oral naming, pointing/matching, simple sorting, counting or drawing.
Questions	What do you notice about the maize? Which plant part are we looking at? What colour, shape or texture can you see?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?